IMMANUEL RAYNALDO SANTJOKO

isantjoko@uh.edu ⋄ https://github.com/NoelsAdventure

A passionate tinkerer and engineer who loves to design, build, and play with robots. Experienced in the design and development of robotics, industrial automation, and embedded systems. Dedicated to pushing the boundaries of technology through theoretical examination, hands-on experimentation, and innovation.

EDUCATION

PhD of Electrical Engineneering Aug. 2025 - Present
University of Houston, USA.

Bachelor of Engineering in Electrical (Mechatronics) Engineering Sep. 2020 - Feb. 2025
Parahyangan Catholic University, Indonesia. GPA : 3.85/4.00

PUBLICATION

T. A. Tamba, I. R. Santjoko, Y. Y. Nazaruddin and V. Nadhira, "A multi-UAV coordination scheme for tracking control of multiple moving target objects," *Proc. IEEE International Conference on Smart Mechatronics*, Yogyakarta, Indonesia, 2024, pp. 7-11. DOI: 10.1109/ICSMech62936.2024.10812277

I. R. Santjoko, T. A. Tamba and A. Sadiyoko, "Modeling and cascade PID controller design of a spinbath circulation process," *Proc. FORTEI-International Conference on Electrical Engineering*, Badung, Indonesia, 2024, pp. 183-188. DOI: 10.1109/FORTEI-ICEE64706.2024.10824351

RECOGNITION

Krisyadana Scholarship for Academic Achievement, Parahyangan Catholic University (2022 - 2024)
Best Graduate of Mechatronics Department, Parahyangan Catholic University (Feb. 2025)

WORK EXPERIENCE

Research Assistant Aug 2025 - Present
University of Houston, USA.

- Safety for robot operation

Research Assistant Sep 2024 - July 2025
Parahyangan Catholic University, Indonesia.

- Experimental design of safety-preserving controller for the prototype of a six degrees-of freedom (6-DOF) Stewart robotic platform
- Design and analysis of cooperative simultaneous localization and pursuit (SLAP) algorithm for multi-UAV cooperative control design with application in tracking and pursuit of moving unmanned surface vehicles
- Modeling and cascade controller design for a spinbath circulation process control to optimize the performance of the overall process

Teaching Assistant Sep 2024 - Jul 2025
Parahyangan Catholic University, Indonesia.
Assisting Engineering Mathematics and Automatic Control Systems courses

Laboratory Assistant Sep 2022 - Jul 2023, Sep 2024 - Jan 2025
Parahyangan Catholic University, Indonesia.
Develop modules and supervise Mechatronics laboratory experiments for such courses as Electronic Systems, Embedded and Microprocessors Systems, and Automatic Control Systems

Automation Engineer Intern Feb 2024 - Jun 2024
Asia Pacific Rayon, Indonesia.
Learned how to design a decentralized control system (DCS), perform maintenance, analyze and resolve fluctuation issues in the spinbath liquid flow, and conducting research on Spinbath Circulation Automation. Made a simulation of cascade PID control system to automate spinbath circulation process based on identification system model.

ORGANIZATION EXPERIENCES

MAHITALA (Hiker and Nature Lover Student Club)
Chief Organizer of the Diving Expedition, Bokan Kepulauan, Indonesia May 2023 - Sep 2023
Disaster Response Volunteer, Cianjur, Indonesia Nov 2022

HMPSTEM (Mechatronics Engineering Student Association)
Secretary and Treasurer Jan 2023 - Des 2023

DEVELOPMENT PROJECTS

Stewart Platform, a 6-DOF parallel manipulator 2025 - Present
Cooperative SLAP, a multi-tracker system to track & pursuit moving targets using range sensor2024 - Present
Spinbath Circulation Automation, a simulation of cascade PID control system to automate spinbath circulation process based on identification system model 2024
Mecanum Robot, a mobile robot that uses a combination of directional forces to achieve omnidirectional movement 2023
FPV Drone, a remotely piloted aircraft system that provides a real-time video feed from an onboard camera to the pilot 2022
Sumo Robot, a remotely controlled combat robot designed to compete in sumo-style robot competitions 2021
Electrical Wheelchair, a device designed to assist individuals with limited mobility 2020

REFERENCE

Tua Agustinus Tamba, Ph.D.
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Department of Electrical Engineering, Parahyangan Catholic University
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